Sampling: from independent draws to Markov chains

Teacher's guide and worked solutions

Bayesian machine learning | Skoltech

Sampling methods turn integrals into averages. Their outputs can look similar while having different mathematical meanings: rejection sampling gives independent target draws, importance sampling gives weighted proposal draws, and Markov chain Monte Carlo gives dependent states with a chosen stationary distribution. This guide develops the theory needed to implement and compare these methods, then gives solutions to three main tasks and an appendix task.

The implementations use NumPy and SciPy. A single parameter and a tractable posterior make it possible to check every method against exact integrals. The same principles extend to models whose posterior normalizer and moments are unavailable. The numerical experiments should be read as controlled examples of the theory, with attention to the distinction between a correct algorithm and a favorable random realization.

1 Monte Carlo integration

1.1 Why Bayesian inference leads to expectations

Let D denote observed data and let θ be a model parameter. Bayes' rule is

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta)p(\theta)\, d\theta.$$

The denominator makes the posterior integrate to one. Evaluating the numerator at a given parameter is often much easier than integrating it over the whole parameter space.

For a future observation y_* with input x_*, conditional independence of the future response and training data given the parameter gives

$$p(y_* \mid x_*, D) = \int p(y_* \mid x_*, \theta)p(\theta \mid D)\, d\theta = \mathbb{E}_{p(\theta|D)}[p(y_* \mid x_*, \theta)].$$

The parameter is integrated out. It should therefore not remain among the conditioning arguments on the left. If posterior draws are available, we average the conditional probabilities evaluated at those draws. Posterior means, event probabilities, and expected losses are instances of the same problem:

$$I = \int f(x)\pi(x)\, dx = \mathbb{E}_\pi[f(X)].$$

An event probability uses an indicator, $f(x) = \mathbf{1}\{x \in A\}$. A predictive probability uses a conditional probability as f. The distribution π and function f play separate roles, and the difficulty of estimating I depends on both.

1.2 Unbiasedness, consistency, and variance

Suppose $X_1, \ldots, X_N$ are independent and identically distributed with density π. The Monte Carlo estimator is

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^{N} f(X_i).$$

If $\mathbb{E}_\pi|f(X)| < \infty$, linearity of expectation gives

$$\mathbb{E}[\widehat{I}_N] = \frac{1}{N}\sum_{i=1}^{N} \mathbb{E}[f(X_i)] = I.$$

Thus it is unbiased at each sample size. Unbiasedness is an average over repeated realizations of the whole experiment. It does not say that a particular estimate is exact, or that its error decreases monotonically as more samples are added.

The strong law of large numbers gives

$$\widehat{I}_N \longrightarrow I \quad \text{almost surely}$$

under the iid and integrability assumptions above. This is a consistency statement. When $\sigma_f^2 = \mathrm{Var}_\pi f(X) < \infty$, independence also gives an explicit finite-sample variance:

$$\begin{aligned}\mathrm{Var}(\widehat{I}_N) &= \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\mathrm{Cov}(f(X_i), f(X_j)) \\ &= \frac{1}{N^2}\sum_{i=1}^{N}\sigma_f^2 = \frac{\sigma_f^2}{N}.\end{aligned}$$

The off-diagonal covariances vanish because the draws are independent. Chebyshev's inequality provides a direct finite-variance route to convergence in probability:

$$\Pr(|\widehat{I}_N - I| > \varepsilon) \le \frac{\sigma_f^2}{N\varepsilon^2} \longrightarrow 0.$$

The bound is often conservative, but it makes the source of the $N^{-1/2}$ error scale explicit.

The root mean squared error is $\mathrm{RMSE}(\widehat{I}_N) = \sqrt{\mathbb{E}[(\widehat{I}_N - I)^2]}$. For any square-integrable estimator, expanding around its expectation gives

$$\mathbb{E}[(\widehat{I}_N - I)^2] = \mathrm{Var}(\widehat{I}_N) + (\mathbb{E}\widehat{I}_N - I)^2.$$

For iid Monte Carlo the bias term vanishes, so RMSE is $\sigma_f/\sqrt{N}$. For a biased estimator such as self-normalized importance sampling, both terms matter. An empirical RMSE from independent replications estimates their combined effect.

1.3 The central limit theorem and Monte Carlo standard error

For iid draws with $0 < \sigma_f^2 < \infty$, the central limit theorem states

$$\frac{\sqrt{N}(\widehat{I}_N - I)}{\sigma_f} \xrightarrow{d} \mathcal{N}(0,1).$$

It is a limiting statement about the estimator's error distribution, not an assertion that a finite-sample average is exactly Gaussian. Replacing σ_f^2 with the sample variance gives

$$s_f^2 = \frac{1}{N-1}\sum_i (f(X_i) - \widehat{I}_N)^2, \qquad \widehat{\mathrm{MCSE}}(\widehat{I}_N) = \frac{s_f}{\sqrt{N}}.$$

An interval such as $\widehat{I}_N \pm 1.96\,\widehat{\mathrm{MCSE}}$ has approximate repeated-experiment coverage when the normal approximation is adequate. For rare events, too few observed events can make this approximation poor. Observing zero events does not establish that the true probability is zero.

The standard deviation σ_f describes variation in the integrand under the target. MCSE describes variation in the numerical estimate of its expectation. For $f(\theta) = \theta$, a posterior interval describes uncertainty about the parameter under the model; an MCSE interval describes numerical error in computing its posterior mean. More draws can shrink the second while leaving the first unchanged.

The exponent in the iid error rate does not explicitly depend on dimension. The variance constant $\mathrm{Var}_\pi f(X)$ can nevertheless grow with dimension, and obtaining useful draws can become much harder. Four times as many independent draws approximately halves MCSE. This concerns the distribution of error; one run's error can increase after adding observations.

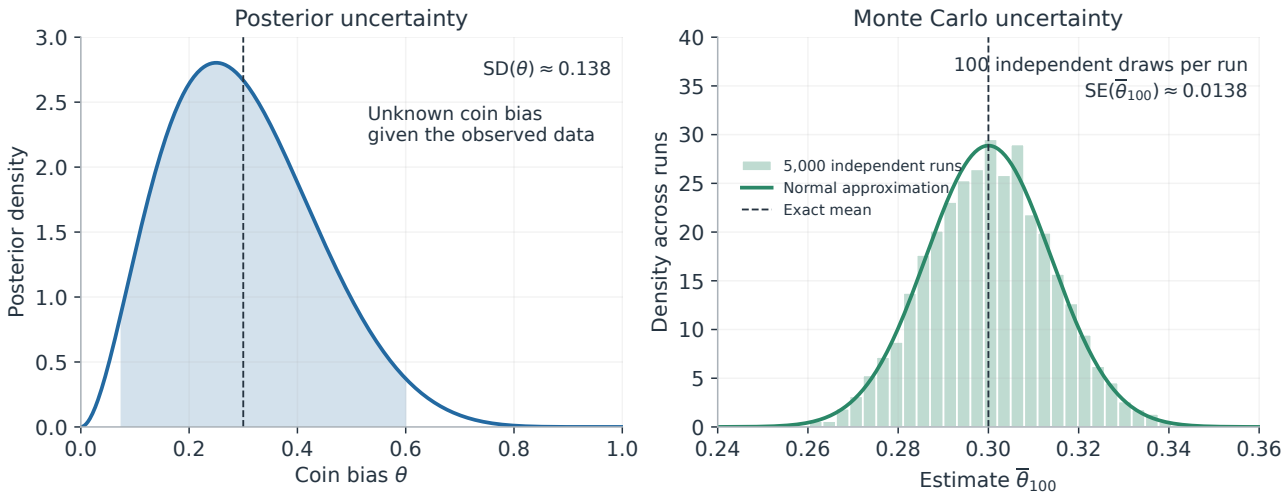


Figure 1: Posterior draws from Beta$(3, 7)$ and estimates of its mean from 100 independent draws describe different random quantities. The panels use different horizontal scales. Increasing the draw count can concentrate the estimates while leaving the posterior unchanged.

1.4 Inverse CDF sampling

Inverse transform sampling converts uniform randomness into draws from a specified distribution. Suppose first that a continuous CDF F is strictly increasing on its support. For $U \sim U(0, 1)$, let $X = F^{-1}(U)$. Then

$$\Pr(X \le x) = \Pr(F^{-1}(U) \le x) = \Pr(U \le F(x)) = F(x).$$

Hence X has CDF F. For an arbitrary CDF, the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \ge u\}$ gives the corresponding construction even when an ordinary inverse does not exist. The claim that $F(X)$ is uniform, in contrast, needs continuity; atoms require care.

For an exponential distribution with rate $\lambda = 2$,

$$F(x) = 1 - e^{-2x}, \quad x \ge 0, \qquad X = -\frac{1}{2}\log(1 - U), \qquad \mathbb{E}X = \frac{1}{2}, \quad \operatorname{Var} X = \frac{1}{4}.$$

The rate is the reciprocal of the scale. The stable numerical expression is `-np.log1p(-u) / 2.0`, which avoids loss of relative accuracy when computing $\log(1 - u)$ for small u.

This example separates a sampling rule from the Monte Carlo estimator applied to its output. Once draws have the correct distribution, the same sample can estimate many expectations. In more complex models the inverse CDF is unavailable, motivating the three tasks below.

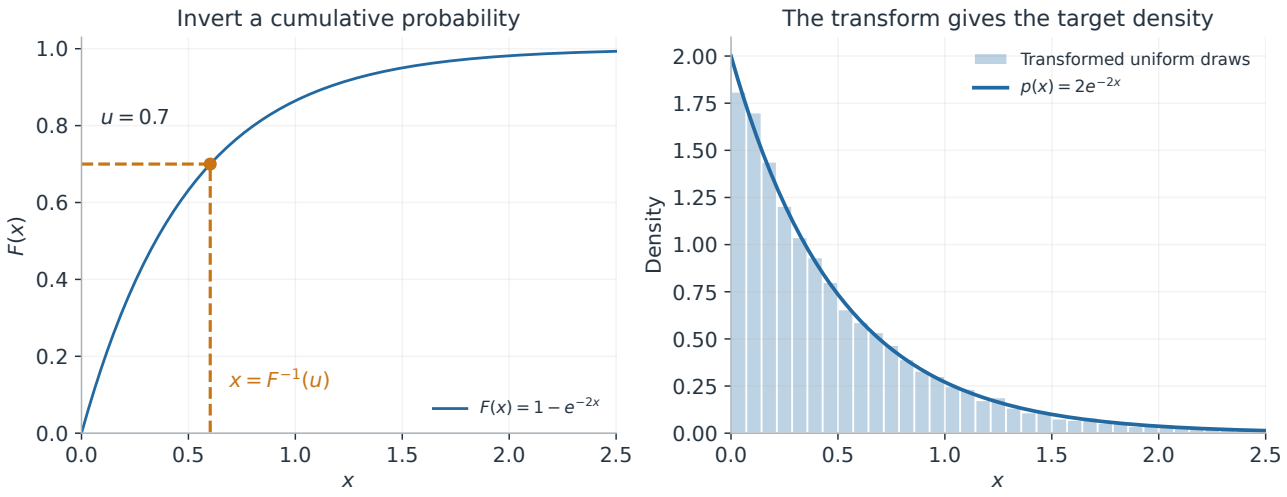


Figure 2: Inverse transform sampling reads a uniform height on the CDF axis and maps it to a location on the sample axis. The exponential distribution has an analytic inverse, so no rejection or Markov chain is needed.

2 A posterior with an exact reference

2.1 The beta–Bernoulli model

Let $\theta \in (0, 1)$ denote a coin's probability of heads. Conditional on θ, the observed tosses are independent Bernoulli variables. If there are h heads and t tails in an observed sequence, its likelihood is

$$p(D \mid \theta) = \theta^h (1 - \theta)^t.$$

If the observation is instead an unordered head count, a binomial coefficient multiplies this expression. That coefficient is independent of θ, so the posterior kernel is unchanged.

For positive shape parameters a_0, b_0, the beta density is

$$p(\theta) = \frac{\theta^{a_0 - 1}(1 - \theta)^{b_0 - 1}}{B(a_0, b_0)}, \qquad B(a, b) = \int_0^1 u^{a-1}(1 - u)^{b-1}\, du.$$

Multiplication by the likelihood adds observed counts to the exponents:

$$p(\theta \mid D) \propto \theta^{a_0 + h - 1}(1 - \theta)^{b_0 + t - 1}.$$

A beta prior is therefore conjugate to Bernoulli observations, with posterior $\mathrm{Beta}(a_0 + h, b_0 + t)$.

Here $a_0 = b_0 = 2$, $h = 1$, and $t = 5$. We choose the unnormalized kernel

$$\widetilde{\pi}(\theta) = \theta^2 (1 - \theta)^6 \mathbf{1}\{0 < \theta < 1\}, \qquad \pi(\theta) = \frac{\widetilde{\pi}(\theta)}{Z} = \mathrm{Beta}(\theta; 3, 7).$$

The constant Z normalizes this chosen kernel. Since constants from the prior density and likelihood may have been omitted, Z is not automatically the model's marginal likelihood. For integer beta parameters,

$$Z = B(3, 7) = \frac{\Gamma(3)\Gamma(7)}{\Gamma(10)} = \frac{2!\, 6!}{9!} = \frac{1}{252}.$$

The tasks operate on $\widetilde{\pi}$ without calling a direct $\mathrm{Beta}(3, 7)$ sampler. The normalized density serves as a reference for auditing the algorithms.

2.2 Exact moments and a posterior event probability

For $\Theta \sim \mathrm{Beta}(a, b)$ and nonnegative integer k, direct integration gives

$$\mathbb{E}\Theta^k = \frac{B(a+k, b)}{B(a, b)}.$$

Using $B(a+1, b) = aB(a, b)/(a+b)$ yields

$$\mathbb{E}\Theta = \frac{a}{a+b}, \qquad \mathbb{E}\Theta^2 = \frac{a(a+1)}{(a+b)(a+b+1)}, \qquad \operatorname{Var}\Theta = \frac{ab}{(a+b)^2(a+b+1)}.$$

Consequently, the two shared estimands and posterior variance are

$$I_1 = \mathbb{E}_\pi\Theta = 0.3, \qquad \operatorname{Var}_\pi\Theta = \frac{21}{1100} \simeq 0.0190909, \qquad I_2 = \Pr_\pi(\Theta > 1/2) = \frac{23}{256} = 0.08984375.$$

The mean is also the posterior predictive probability of a head on the next toss because $\Pr(H_{\text{next}} \mid \Theta) = \Theta$.

The tail reference can be derived without simulation. An antiderivative of $\theta^2(1-\theta)^6$ is

$$A(\theta) = \frac{\theta^3}{3} - \frac{6\theta^4}{4} + \frac{15\theta^5}{5} - \frac{20\theta^6}{6} + \frac{15\theta^7}{7} - \frac{6\theta^8}{8} + \frac{\theta^9}{9}.$$

Therefore $I_2 = 252[A(1) - A(1/2)] = 23/256$. Equivalently, the integer-shape beta–binomial identity gives $I_2 = 2^{-9}\sum_{k=0}^{2}\binom{9}{k} = (1+9+36)/512$. The event probability is an expectation of $f(\theta) = \mathbf{1}\{\theta > 1/2\}$, so it needs no new sampling rule.

2.3 Evaluating the target in log space

Inside the support, $\log\widetilde{\pi}(\theta) = 2\log\theta + 6\log(1-\theta)$. Outside it, the log kernel is $-\infty$. The helper below preserves valid interior arguments while avoiding logarithms of invalid ones:

```
import numpy as np
from scipy.special import logsumexp
from scipy.stats import norm

def log_target(theta):
    theta = np.asarray(theta, dtype=float)
    inside = (theta > 0.0) & (theta < 1.0)
    safe = np.where(inside, theta, 0.5)
    value = 2*np.log(safe) + 6*np.log1p(-safe)
    return np.where(inside, value, -np.inf)
```

Replacing out-of-support arguments temporarily only prevents invalid arithmetic; the final result still assigns them zero density. Clipping valid interior arguments to an arbitrary tolerance would change the declared target near its boundaries and is unnecessary.

3 Task 1: rejection sampling

3.1 The envelope and the augmented joint law

Suppose q is a normalized density from which independent proposals can be drawn, and a finite constant M satisfies

$$0 \le \widetilde{\pi}(x) \le Mq(x) \quad \text{for every } x.$$

This requires q to cover the target support and to be large enough relative to the kernel everywhere. A proposal with lighter tails than the target may admit no finite envelope even if it resembles the target near its mode.

Draw $Y \sim q$ and $U \sim U(0,1)$ independently, and retain Y on the event

$$A = \left\{ U < \frac{\tilde{\pi}(Y)}{Mq(Y)} \right\}.$$

The joint density of (Y, U) is $q(y)$ on $0 < u < 1$. Integrating over the accepted vertical segment gives

$$p(Y \in dy, A) = q(y) \frac{\tilde{\pi}(y)}{Mq(y)}\, dy = \frac{\tilde{\pi}(y)}{M}\, dy.$$

Integrating once more yields the acceptance probability

$$a = \Pr(A) = \frac{1}{M} \int \tilde{\pi}(y)\, dy = \frac{Z}{M}.$$

Dividing the accepted joint mass by a shows that

$$p(y \mid A) = \frac{\tilde{\pi}(y)/M}{Z/M} = \pi(y).$$

An alternative geometric construction draws a height $T \mid Y = y \sim U(0, Mq(y))$. Its joint density with Y is $1/M$ under the envelope. Conditioning on $0 < T < \tilde{\pi}(Y)$ gives the target marginal. The area under the kernel divided by the area under the envelope is Z/M.

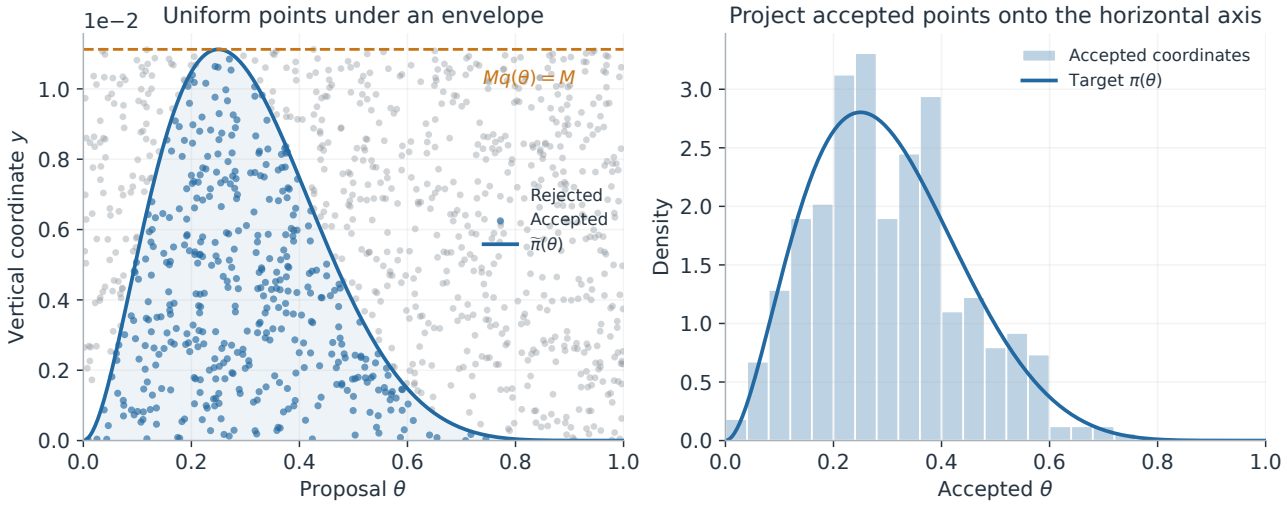


Figure 3: Rejection sampling augments each proposal with an independent height. Keeping points below the target kernel gives a horizontal marginal proportional to that kernel. The unused part of the envelope determines rejection cost.

The acceptance formula $1/M$ applies only when the target in the envelope is normalized. Here M bounds an unnormalized kernel and can be less than one. Multiplying the kernel and envelope by the same positive constant changes neither acceptance decisions nor the conditional law.

3.2 A uniform envelope for the coin posterior

Use $q(\theta) = 1$ on $(0,1)$. Maximizing the log kernel gives

$$\frac{d}{d\theta} \log \tilde{\pi}(\theta) = \frac{2}{\theta} - \frac{6}{1-\theta} = 0 \quad \Longrightarrow \quad \theta_{\text{mode}} = \frac{1}{4}.$$

Its second derivative is $-2/\theta^2 - 6/(1-\theta)^2 < 0$, so this is the unique interior maximum. The kernel tends to zero at both boundaries. A tight envelope is therefore

$$M = \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^6 = \frac{729}{65536} \simeq 0.01112366, \qquad a = \frac{Z}{M} \simeq 0.35674004.$$

An underestimated envelope makes some nominal acceptance probabilities exceed one. Clipping them to one does not repair the derivation: the accepted density is then proportional to $\min\{\tilde{\pi}, Mq\}$ rather than the target kernel. A larger valid envelope preserves correctness and reduces efficiency.

3.3 Implementation and output contract

The function `rejection_sample` takes `n_proposals` and `rng`. It returns accepted values and a Boolean mask with one entry per proposal. The proposal budget is fixed at 20,000; the retained count is random. Candidate uniforms and acceptance uniforms must be independent.

```
def rejection_sample(n_proposals, rng):
    candidates = rng.uniform(0.0, 1.0, size=n_proposals)
    uniforms = rng.uniform(0.0, 1.0, size=n_proposals)
    envelope = (1.0 / 4.0)**2 * (3.0 / 4.0)**6
    kernel = candidates**2 * (1.0 - candidates)**6
    mask = uniforms < kernel / envelope
    return candidates[mask], mask
```

For independent proposal–uniform pairs, accepted locations are independent target draws. Conditional on the accepted count $K = k > 0$ under a fixed proposal budget, retained locations have the iid target law. Their sample-based Monte Carlo standard error therefore uses k:

```
rng = np.random.default_rng(11)
accepted, mask = rejection_sample(20_000, rng)
mean_estimate = accepted.mean()
mean_se = accepted.std(ddof=1) / np.sqrt(len(accepted))
tail_values = (accepted > 0.5).astype(float)
tail_estimate = tail_values.mean()
tail_se = tail_values.std(ddof=1) / np.sqrt(len(accepted))
acceptance_estimate = mask.mean()
```

At least two accepted values are needed for a sample variance with `ddof=1`. A general implementation should report insufficient samples when that condition fails.

3.4 Model answers and useful hints

The count K is binomial with mean $20{,}000a \simeq 7{,}135$ and standard deviation $\sqrt{20{,}000a(1-a)} \simeq 67.7$. Its empirical acceptance rate has standard deviation about 0.00339. For a retained count near its mean, the posterior mean's theoretical MCSE is approximately 0.00164, while the tail estimate's is approximately 0.00339. These are error scales, not fixed outputs or guarantees for a particular random seed.

A useful first hint is to construct two arrays of the same length. The next hint is to evaluate the kernel at all candidates and divide by the envelope. The final step is a Boolean comparison followed by `candidates[mask]`. Returning indices instead of a mask changes the interface.

The posterior standard deviation is approximately 0.13817, much larger than the MCSE of its estimated mean. A density histogram of accepted values estimates the posterior; a histogram of all candidates estimates the proposal. Keeping rejected candidates would mix two different sampling laws.

The method needs about $1/a \simeq 2.8$ proposals per retained draw on average. This describes proposal cost, rather than a complete comparison with other methods: evaluating a proposal, its density, and the target can have different costs in different models.

4 Task 2: self-normalized importance sampling

4.1 Changing the sampling distribution

Importance sampling changes the distribution generating locations and compensates with a density ratio. Suppose the normalized target π is known, $Y_i \sim q$ independently, and $q(x) > 0$ wherever

$|f(x)|\pi(x) > 0$. Then

$$I = \int f(x)\pi(x)\,dx = \int f(x)\frac{\pi(x)}{q(x)}q(x)\,dx = \mathbb{E}_q\left[\frac{\pi(Y)}{q(Y)}f(Y)\right].$$

The ordinary importance estimator is

$$\widehat{I}_{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N}\frac{\pi(Y_i)}{q(Y_i)}f(Y_i).$$

If the absolute expectation exists, it is unbiased. If its second moment is finite, its variance is

$$\mathrm{Var}(\widehat{I}_{\mathrm{IS}}) = \frac{1}{N}\left[\int \frac{\pi(x)^2 f(x)^2}{q(x)}\,dx - I^2\right].$$

Small q where $|f|\pi$ is appreciable can make this variance very large or infinite. Sampling from another distribution is useful only when its importance correction remains manageable.

For several estimands, requiring $q > 0$ on the whole target support is a convenient condition. It is essential for the self-normalized construction below, whose denominator must estimate the entire target mass.

4.2 The ratio estimator when the normalizer is unknown

Let $\pi = \widetilde{\pi}/Z$ and define $w(x) = \widetilde{\pi}(x)/q(x)$. Coverage of the target support gives

$$\mathbb{E}_q[w(Y)] = Z, \qquad \mathbb{E}_q[w(Y)f(Y)] = ZI.$$

Estimate both expectations from the same draws:

$$\widehat{I}_{\mathrm{SNIS}} = \frac{N^{-1}\sum_i w_i f(Y_i)}{N^{-1}\sum_i w_i} = \sum_i \bar{w}_i f(Y_i), \qquad \bar{w}_i = \frac{w_i}{\sum_j w_j}.$$

The unknown Z cancels. The locations still have law q: weighting changes the estimator, not their sampling distribution. Resampling from the weighted empirical distribution is a separate operation and is unnecessary for these expectations.

The expectation of a ratio is generally not the ratio of expectations. SNIS is therefore generally biased at finite N. A constant integrand is a special case with zero bias, not a proof of general unbiasedness. With $0 < Z < \infty$, coverage, and $\mathbb{E}_\pi|f| < \infty$, the laws of large numbers for numerator and denominator imply almost sure consistency wherever the ratio is defined. A positive weight sum is required; proposals entirely outside target support cannot produce an estimate.

4.3 Asymptotic variance and a plug-in MCSE

The centered ratio has the exact representation

$$\sqrt{N}(\widehat{I}_{\mathrm{SNIS}} - I) = \frac{N^{-1/2}\sum_i w_i(f(Y_i) - I)}{N^{-1}\sum_i w_i}.$$

The numerator summands have mean zero. If $\mathbb{E}_q[w^2(f-I)^2] < \infty$, their iid central limit theorem and the denominator's law of large numbers give

$$\sqrt{N}(\widehat{I}_{\mathrm{SNIS}} - I) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\mathrm{SNIS,f}}), \qquad \sigma^2_{\mathrm{SNIS,f}} = \frac{\mathbb{E}_q[w^2(f-I)^2]}{Z^2}.$$

The variance depends on f. A proposal that estimates the mean well need not be best for an event probability.

Replacing the unknown terms by their sample counterparts yields

$$\widehat{\mathrm{MCSE}}_{\mathrm{SNIS}} = \sqrt{\sum_i \bar{w}_i^2 (f(Y_i) - \widehat{I}_{\mathrm{SNIS}})^2}.$$

The squared expression is the plug-in estimate of $\sigma^2_{\mathrm{SNIS,f}}/N$. It is a large-sample estimate under iid covered proposal sampling and a finite weighted second moment. It is not an unbiased finite-sample variance estimator or a remedy for an unvisited important region. Compute it separately for the mean and tail indicator.

The ratio estimator and importance-sampling error analysis are developed in Owen's treatment [1]. The practical questions are whether the proposal covers the target and whether the relevant weighted moments are finite.

4.4 Stable normalized weights and weight ESS

Raw importance weights can overflow or underflow. Writing $\ell_i = \log w_i$ and $a = \max_i \ell_i$ gives

$$v_i = e^{\ell_i - a}, \qquad \bar{w}_i = \frac{v_i}{\sum_j v_j}.$$

The common factor e^{-a} cancels. Equivalently, $\bar{w}_i = \exp(\ell_i - \mathrm{logsumexp}(\ell))$. Adding a common finite constant to all log weights leaves the normalized weights, estimates, and weight ESS unchanged.

The weight concentration diagnostic is

$$\mathrm{ESS}_w = \frac{1}{\sum_i \bar{w}_i^2} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

Because $\sum_i \bar{w}_i = 1$, Cauchy–Schwarz gives $1 \le N \sum_i \bar{w}_i^2$, so $\mathrm{ESS}_w \le N$. Nonnegativity implies $\sum_i \bar{w}_i^2 \le (\sum_i \bar{w}_i)^2 = 1$, giving $\mathrm{ESS}_w \ge 1$. Equal weights achieve N; concentration on one point gives one.

If $\mathbb{E}_q w^2 < \infty$, a law of large numbers gives

$$\frac{\mathrm{ESS}_w}{N} = \frac{(N^{-1}\sum_i w_i)^2}{N^{-1}\sum_i w_i^2} \longrightarrow \frac{Z^2}{\mathbb{E}_q w^2} = \frac{1}{1 + \mathrm{Var}_q(w)/Z^2}.$$

Thus weight ESS measures weight dispersion. It is not an exact equivalent count of target draws for every integrand: its formula contains no f, while the SNIS asymptotic variance does. It also differs from the autocorrelation MCMC ESS derived later.

4.5 Implementation and the proposal comparison

The function `snis` takes matching one-dimensional arrays of log weights and finite values. At least one log weight is finite; others may be $-\infty$ to represent zero weight. NaN and positive infinity are excluded. It returns the estimate, normalized weights, and scalar weight ESS.

```
def snis(log_weights, values):
    log_weights = np.asarray(log_weights, dtype=float)
    values = np.asarray(values, dtype=float)
    maximum = np.max(log_weights)
    weights = np.exp(log_weights - maximum)
    normalized_weights = weights / weights.sum()
    estimate = np.sum(normalized_weights * values)
    ess = 1.0 / np.sum(normalized_weights**2)
    return estimate, normalized_weights, ess
```

The returned weights also support weighted plots and the plug-in MCSE:

```
estimate, normalized_weights, ess = snis(log_weights, values)
mcse = np.sqrt(np.sum(normalized_weights**2 * (values - estimate)**2))
```

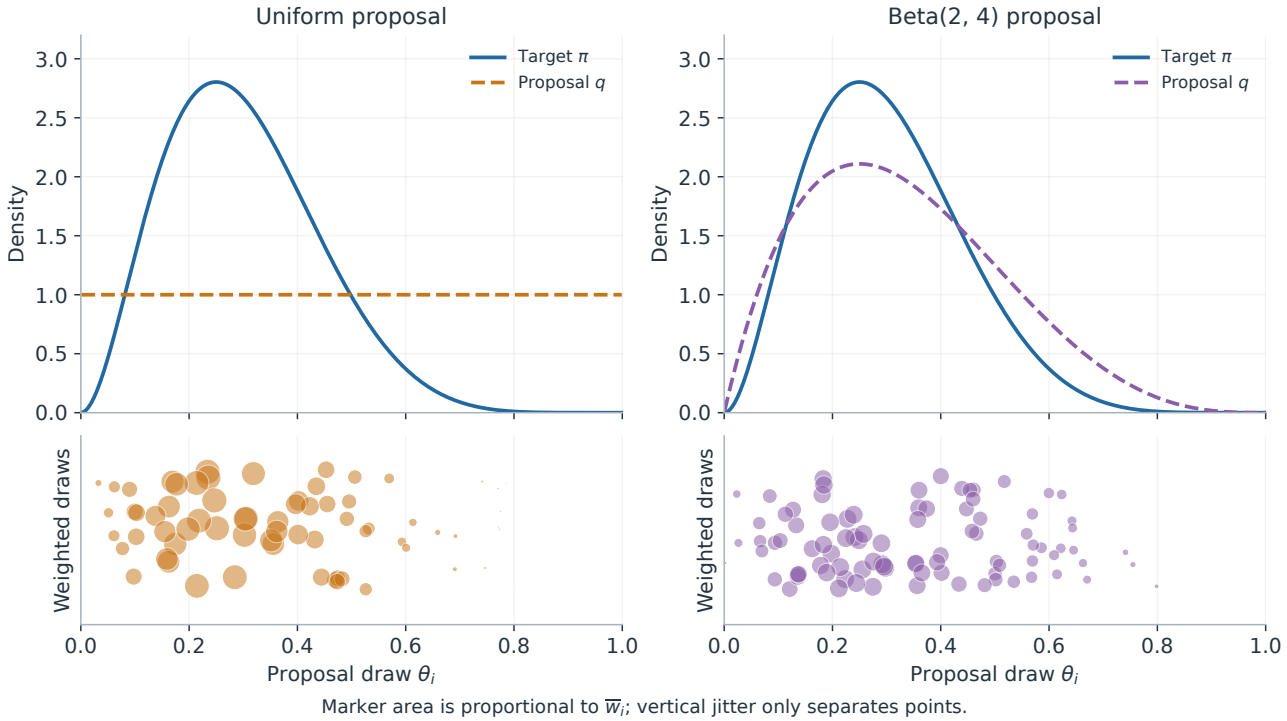


Figure 4: Importance sampling retains proposal locations and changes their contributions to an estimate. Coverage and weight concentration both matter; a weighted empirical distribution differs from an unweighted sample drawn directly from the target.

Reuse locations and weights for both quantities. Use x as the values for the mean and the indicator $\mathbf{1}\{x > 0.5\}$ for the tail. An unweighted proposal mean estimates an expectation under q. An average of unnormalized `weights * values` estimates ZI and cannot replace the ratio when Z is unknown.

The comparison uses $q_U = U(0,1)$ and $q_B = \text{Beta}(2,4)$, with 2,000 proposals each. Under the uniform proposal, $w_U = \tilde{\pi}$. Since $B(2,4) = 1/20$,

$$q_B(\theta) = 20\theta(1-\theta)^3, \qquad w_B(\theta) = \frac{\theta(1-\theta)^3}{20}.$$

Both proposals cover $(0,1)$ and have finite relevant second moments. The beta proposal concentrates more mass in the posterior's high-density region and typically gives more balanced weights.

For these proposals,

$$\mathbb{E}_{q_U} w^2 = B(5,13), \qquad \mathbb{E}_{q_B} w^2 = \frac{B(4,10)}{20}.$$

The limiting ESS fractions are approximately 0.487 and 0.901. At 2,000 draws, weight ESS values around 975 and 1,800 are reasonable scales, not requirements for every realization.

4.6 Model answers, repeated estimates, and hints

The repeated experiment uses 100 independent replications with 500 proposals each. For estimates $\widehat{I}_r$, empirical bias and RMSE are

$$\widehat{\text{bias}} = \frac{1}{R}\sum_r \widehat{I}_r - I, \qquad \widehat{\text{RMSE}} = \sqrt{\frac{1}{R}\sum_r (\widehat{I}_r - I)^2}.$$

The mean's asymptotic error scales at this proposal count are approximately 0.0067 for uniform proposals and 0.0055 for beta proposals. For the tail, they are closer: approximately 0.0108 and 0.0101. Finite replication error can reverse small differences. A small empirical bias does not establish finite-sample unbiasedness.

A useful sequence of hints is to normalize log weights, form a weighted sum, then compute the squared-weight sum. Equal weights should give the arithmetic mean and ESS N. A single nonzero weight gives its location's value and ESS one. Adding 1000 to every log weight tests stability and invariance to a target normalizer.

The explanation should distinguish increased weight ESS from a universal accuracy claim. A proposal can miss an important region and still produce apparently even weights among observed points. Neither large observed ESS nor a narrow plug-in MCSE establishes adequate coverage from observations alone. Both rely on proposal and moment assumptions.

5 Task 3: random-walk Metropolis–Hastings

5.1 A Markov chain and its stationary law

A Markov chain has conditional independence

$$p(x_{t+1} \mid x_0, \ldots, x_t) = p(x_{t+1} \mid x_t).$$

A time-homogeneous kernel $K(x, A)$ gives the probability of moving from x into a measurable set A, using the same rule at every iteration. A distribution π is stationary for K if

$$\pi(A) = \int K(x, A)\pi(x)\, dx$$

for every A. Starting a chain in π preserves that marginal law after each transition. Starting elsewhere need not produce that law immediately.

MCMC builds a usable kernel with the desired stationary distribution. Stationarity is invariance; convergence from other starts also needs suitable irreducibility, recurrence, and aperiodicity conditions. Even an ergodic chain can mix too slowly for a practical run. Finite-run diagnostics assess consequences of this difficulty without proving the absence of unseen problems.

5.2 Acceptance and the full transition kernel

From current state x with positive target density, propose $Y \sim q(\cdot \mid x)$ and accept with probability

$$\alpha(x, y) = \min\left\{1, \frac{\widetilde{\pi}(y)q(x \mid y)}{\widetilde{\pi}(x)q(y \mid x)}\right\}.$$

A zero reverse proposal density or zero candidate target density gives acceptance zero. The unknown target normalizer cancels in the ratio.

The transition includes a continuous move and a point mass at the current state:

$$K(x, dy) = q(y \mid x)\alpha(x, y)\, dy + r(x)\delta_x(dy), \qquad r(x) = 1 - \int q(z \mid x)\alpha(x, z)\, dz.$$

Here δ_x puts all mass at x. The continuous accepted-move density usually integrates to less than one. The holding mass completes the transition and is the mathematical reason repeated states must be stored.

The accepted flow satisfies

$$\begin{aligned}\pi(x)q(y \mid x)\alpha(x, y) &= \min\{\pi(x)q(y \mid x), \pi(y)q(x \mid y)\} \\ &= \pi(y)q(x \mid y)\alpha(y, x).\end{aligned}$$

The holding part lies on the diagonal $x = y$ and is symmetric too. Thus $\pi(dx)K(x, dy) = \pi(dy)K(y, dx)$, the detailed balance identity. Integrating over x gives stationarity:

$$\int \pi(dx)K(x, A) = \int_A \pi(dy) \int K(y, dx) = \pi(A).$$

Detailed balance is a sufficient construction of stationarity; it is not a necessary property of every MCMC method.

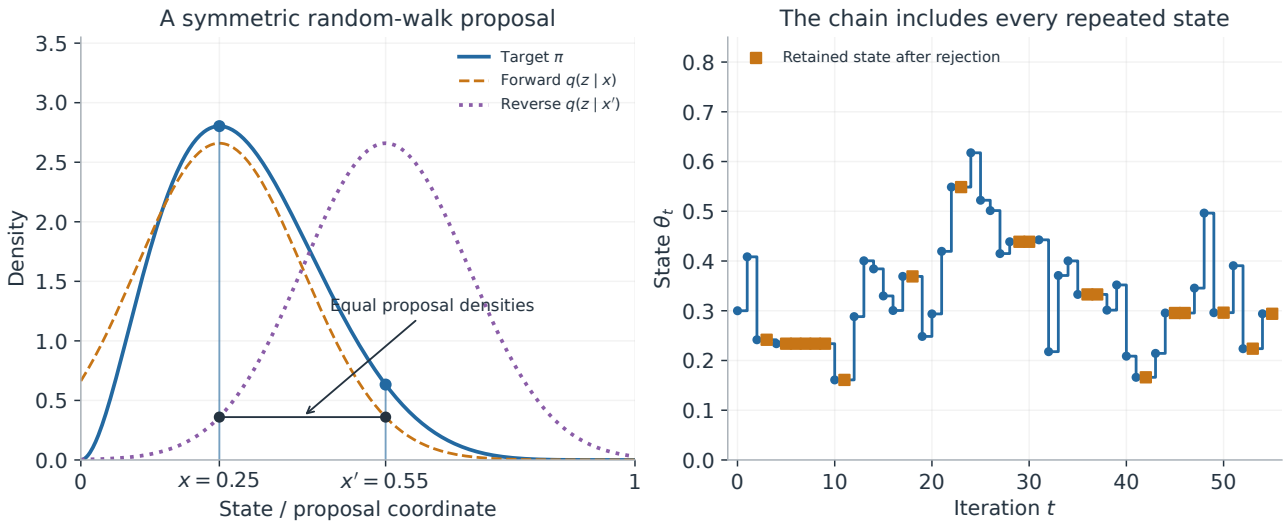


Figure 5: A Metropolis step combines a proposal with an acceptance decision. Rejection is a genuine holding probability. Small proposals can be accepted frequently while moving slowly through the target.

5.3 Symmetric Gaussian proposals and log-space decisions

For the random walk

$$Y = x + s\varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1), \qquad q(y \mid x) = \mathcal{N}(y; x, s^2),$$

forward and reverse densities are equal. Therefore

$$\log \alpha(x, y) = \min\{0, \log \widetilde{\pi}(y) - \log \widetilde{\pi}(x)\}.$$

Accepting when $\log U < \log \alpha$ avoids exponentiating a potentially enormous density ratio. An out-of-support candidate has log target $-\infty$ and is rejected. Initial states must have finite log target.

The Gaussian proposal is sampled on its full support, and outside candidates are rejected. Moving an outside candidate into the target interior changes the accepted proposal kernel. In this example, mapping it exactly to an endpoint and still assigning that endpoint log target $-\infty$ gives the same rejection decision, but serves no purpose. Resampling until a candidate is inside instead produces a truncated proposal whose normalizer depends on the current state. Such a proposal needs its own Hastings correction.

5.4 Implementation and vectorized chains

Return `samples` and `acceptance_rate`, shaped (C, T) and $(C,)$ for C chains and T transitions. Column t stores the state after decision t, including repeats after rejection. Copy the input array.

```
def random_walk_metropolis(log_target, initial, proposal_std,
                           n_steps, rng):
    current = np.asarray(initial, dtype=float).copy()
    log_current = np.asarray(log_target(current), dtype=float)
    n_chains = len(current)
    samples = np.empty((n_chains, n_steps))
    accepted = np.zeros(n_chains, dtype=int)
    for step in range(n_steps):
        candidate = current + rng.normal(
            0.0, proposal_std, size=n_chains)
        log_candidate = log_target(candidate)
        log_alpha = np.minimum(0.0, log_candidate - log_current)
        log_u = np.log(rng.random(n_chains))
        take = log_u < log_alpha
        current[take] = candidate[take]
        log_current[take] = log_candidate[take]
        accepted += take
        samples[:, step] = current
    return samples, accepted / n_steps
```

Independent proposal noise and acceptance uniforms are drawn for each chain and iteration. The chains share vectorized computation but do not exchange states. Caching the current log density avoids re-evaluating an unchanged point after rejection.

A first hint is to represent the current state and log density as arrays of length C. Construct the candidate and Boolean acceptance mask; update both cached arrays on that mask; save the whole current-state array. The saved column count equals the number of transitions, not the accepted count.

5.5 Why deleting repeated states changes the distribution

Let $a(x) = 1 - r(x)$ be the probability of an accepted move from x. If only accepted states are retained, their transition kernel is, where $a(x) > 0$,

$$J(x, dy) = \frac{q(y \mid x)\alpha(x, y)}{a(x)}\, dy.$$

The invariant density of this embedded jump chain is generally proportional to $\pi(x)a(x)$. Multiplying J by $\pi(x)a(x)$ cancels the denominator and recovers the accepted-flow detailed balance identity. States with higher acceptance probability are therefore overrepresented when holding periods are discarded.

This differs from rejection sampling. Rejection sampling makes independent proposals and conditions each on acceptance. MH starts the next proposal from the current state, and its state-dependent holding probability is part of the target-preserving kernel.

5.6 Interpreting the proposal-scale experiment

The comparison uses starts $[0.05, 0.25, 0.65, 0.95]$, 6,000 transitions per chain, and discards the first 1,000 states. Proposal standard deviations are 0.01, 0.15, and 1.0.

At $s = 0.01$ most proposals are accepted, but local moves are small. Traces can retain information about initialization and show strong serial dependence. At $s = 1$ many proposals leave the support or reach low-density locations, so holds are frequent. The intermediate scale typically explores more efficiently.

Acceptance alone is insufficient for tuning. The goal is a precise estimate of the required expectation per computational cost. No universal acceptance percentage follows from this scalar example. Discarding a fixed prefix can reduce transient effects but cannot guarantee stationarity or remove dependence.

6 Monte Carlo error for dependent draws

6.1 Finite-sample variance and integrated autocorrelation

Assume temporarily that the chain is stationary. Let $G_t = f(X_t)$ have mean I and finite variance γ_0. Define $\gamma_k = \mathrm{Cov}(G_t, G_{t+k})$ and $\rho_k = \gamma_k/\gamma_0$ when $\gamma_0 > 0$. Expanding covariances gives the exact identity

$$\begin{aligned}\mathrm{Var}(\widehat{I}_N) &= \frac{1}{N^2}\left[N\gamma_0 + 2\sum_{k=1}^{N-1}(N-k)\gamma_k\right] \\ &= \frac{\gamma_0}{N}\left[1 + 2\sum_{k=1}^{N-1}\left(1-\frac{k}{N}\right)\rho_k\right].\end{aligned}$$

There are $N-k$ pairs at lag k, explaining the triangular weight. The iid formula is recovered when all nonzero-lag covariances vanish.

When the autocovariance sum converges appropriately, define

$$S_f = \gamma_0 + 2\sum_{k=1}^{\infty}\gamma_k, \qquad \tau_f = \frac{S_f}{\gamma_0} = 1 + 2\sum_{k=1}^{\infty}\rho_k.$$

Under suitable Markov-chain central limit theorem conditions, $\sqrt{N}(\widehat{I}_N - I)$ approaches $\mathcal{N}(0, S_f)$, and

$$\mathrm{MCSE}(\widehat{I}_N) \simeq \sqrt{S_f/N}, \qquad \mathrm{ESS}_{\mathrm{MCMC},f} \simeq N/\tau_f.$$

These expressions are asymptotic and assume the intended stationary law. Initial bias is not included in S_f/N.

Dependence is specific to f. Mean and tail-indicator estimates need separate checks. Lag-one correlation describes only local persistence; slow decay at larger lags can dominate the variance. Negative correlation can in principle give ESS larger than N. Capping ESS at N is a reporting convention, not a universal mathematical upper bound.

6.2 Batch-means MCSE for independent chains

Nonoverlapping batches estimate the long-run variance without estimating every autocorrelation. With C independent retained chains, choose batch size b and use $n = ab$ states per chain, trimming the remainder. Define

$$\bar{G}_{cj} = \frac{1}{b}\sum_{t=(j-1)b+1}^{jb} G_{ct}, \quad j = 1, \ldots, a, \qquad \bar{G}_c = \frac{1}{a}\sum_j \bar{G}_{cj}.$$

For sufficiently long batches of stationary draws, a batch mean has variance approximately S_c/b. This motivates

$$\widehat{S}_c = \frac{b}{a-1}\sum_{j=1}^{a}(\bar{G}_{cj} - \bar{G}_c)^2.$$

For an equally weighted pooled mean, independence between chains gives

$$\widehat{\mathrm{MCSE}}^{\,2} = \frac{1}{C^2 n}\sum_{c=1}^{C}\widehat{S}_c = \frac{C^{-1}\sum_c \widehat{S}_c}{Cn}.$$

The experiment uses $b = 100$. Increasing the batch size is a useful sensitivity check while enough batches remain. Short batches can underestimate long-range dependence; too few batches give a noisy variance estimate. Formal consistency requires suitable growing-batch and chain conditions, not a fixed small batch size at every run length.

The approximate batch ESS uses a within-chain marginal variance estimate:

$$\bar{v} = \frac{1}{C}\sum_{c}\frac{1}{n-1}\sum_{t}(G_{ct}-\bar{G}_c)^2, \qquad \widehat{\mathrm{ESS}}_{\mathrm{batch}} = \mathrm{clip}\left(\frac{\bar{v}}{\widehat{\mathrm{MCSE}}^2}, 1, Cn\right).$$

This is a rough diagnostic, not a replacement for modern MCMC ESS estimation. If chains are trapped, batch variance can be small while estimates are wrong. A constant tail-indicator sequence has no observed variation but cannot establish a zero tail probability. Undefined diagnostics must remain explicitly undefined rather than being interpreted as successful convergence.

6.3 Classical split $\widehat{R}$

Several chains allow comparison between within-chain variation and disagreement between chain locations. Split each retained chain in half; if its length is odd, omit the unused middle value. Let $m = 2C$ be the number of split sequences and n_s their length. Write the mean of split sequence j as $\bar{G}_j$ and its sample variance as s_j^2. Define

$$\bar{G} = \frac{1}{m}\sum_{j}\bar{G}_j, \quad W = \frac{1}{m}\sum_{j}s_j^2, \quad B = \frac{n_s}{m-1}\sum_{j}(\bar{G}_j-\bar{G})^2.$$

The classical pooled variance estimate and diagnostic are

$$\widehat{V} = \frac{n_s-1}{n_s}W + \frac{B}{n_s}, \qquad \widehat{R} = \sqrt{\widehat{V}/W}.$$

If chain locations disagree relative to within-chain variation, B inflates the ratio. Splitting can expose differences between earlier and later portions. Large values argue against satisfactory mixing. Finite-sample values may be slightly below one; a value near one does not prove convergence.

If $W = 0$ but $B > 0$, the ratio is infinite. If both vanish, it is undefined. Identical constant chains should not be declared converged. Nonconstant chains can also miss the same region while appearing to agree.

This is the classical variance-based construction. Modern rank-normalized and folded $\widehat{R}$, together with bulk and tail ESS, address important failures of the classical diagnostic [2]. The simplified formulas make the mechanism explicit, but their limitations remain part of the interpretation. Traces, estimand-specific summaries, batch sensitivity, and between-chain agreement provide complementary evidence.

6.4 Model answers for the chain comparison

The small scale can combine high acceptance with persistent traces, high lag correlation, and disagreement between chain means. The large scale can combine low acceptance with many repeated states. The intermediate scale is reasonable when traces explore the posterior and MCSE and between-chain diagnostics support that reading.

The naive iid standard error ignores the off-diagonal covariances. It often understates the error in these positively correlated examples. Exact empirical orderings can fluctuate because the diagnostics are estimated. Thinning is not a general substitute for assessing dependence: discarding states also discards information unless there is a separate storage or downstream cost constraint.

A defensible explanation identifies the chosen scale, cites at least two diagnostics, and recognizes the finite-run limitation. A favorable histogram or one numerical agreement with the reference cannot establish an arbitrary algorithm's correctness.

7 Comparing the three constructions

The tasks estimate the same mean and event probability with different outputs:

Method	Required construction	Meaning of the output
Rejection	A sampleable proposal and valid global envelope	Independent target values after acceptance; the retained count is random.
Importance	A sampleable, evaluable proposal covering the target	Independent proposal locations with weights used in an estimator.
Metropolis–Hastings	An evaluable target kernel and a correctly accounted proposal	Every state of a dependent chain, including repeats after rejection.

Different budgets in the demonstrations prevent their tables from being a controlled performance benchmark. A comparison per target evaluation must account for all proposals, rejected candidates, discarded initial states, and proposal-density costs.

The normalizer is absent from the rejection decision because a kernel is compared with its envelope. SNIS cancels it between numerator and denominator. MH cancels it between target values. None of these cancellations alone guarantees efficiency.

Increasing independent posterior draws reduces numerical error under finite variance. Increasing the importance budget helps under coverage and moment conditions but cannot fix missing support. Extending an ergodic MCMC run can reduce transient and correlated sampling error, while a poor proposal can still make the required run impractically long. Changing a proposal can be more useful than increasing the iteration count.

For vector-valued posterior draws in logistic regression, let $\sigma(a) = 1/(1 + e^{-a})$ and estimate the predictive probability by averaging $\sigma(\theta_0^{(i)} + x_*\theta_1^{(i)})$. Average after applying the nonlinearity, rather than applying it once to the mean parameter. With dependent draws, MCSE concerns this predictive-probability integrand and its own autocorrelation.

Numerical agreement matters only alongside a valid estimator. A wrong algorithm can accidentally reproduce a low-order moment. The appendix gives a clear illustration.

A Task A: asymmetric proposals and multimodality

A.1 Two separated modes

Consider the equal mixture

$$\pi_{\text{mix}}(x) = \tfrac{1}{2}\mathcal{N}(x; -4, 0.6^2) + \tfrac{1}{2}\mathcal{N}(x; 4, 0.6^2).$$

Symmetry gives $\mathbb{E}X = 0$ and $\Pr(X > 0) = 1/2$. Total variance gives

$$\operatorname{Var} X = \mathbb{E}[\operatorname{Var}(X \mid \text{component})] + \operatorname{Var}(\mathbb{E}[X \mid \text{component}]) = 0.36 + 16 = 16.36.$$

Evaluate the log density using log-sum-exp over component log densities, rather than adding log densities or taking a logarithm after underflow:

```
def log_mixture(x):
    components = np.stack([norm.logpdf(x, -4.0, 0.6),
                           norm.logpdf(x, 4.0, 0.6)])
    return logsumexp(components, axis=0) - np.log(2.0)
```

Four chains start at $[-4, -4, 4, 4]$. The comparison uses 8,000 transitions, discards the first 1,000, and contrasts a local Gaussian random walk of standard deviation 0.4 with an independence proposal $q(y \mid x) = q(y) = \mathcal{N}(y; 0, 5^2)$.

A.2 The general Hastings correction

The function takes current and candidate target log densities, then forward and reverse proposal log densities. Its output is log acceptance, capped at zero:

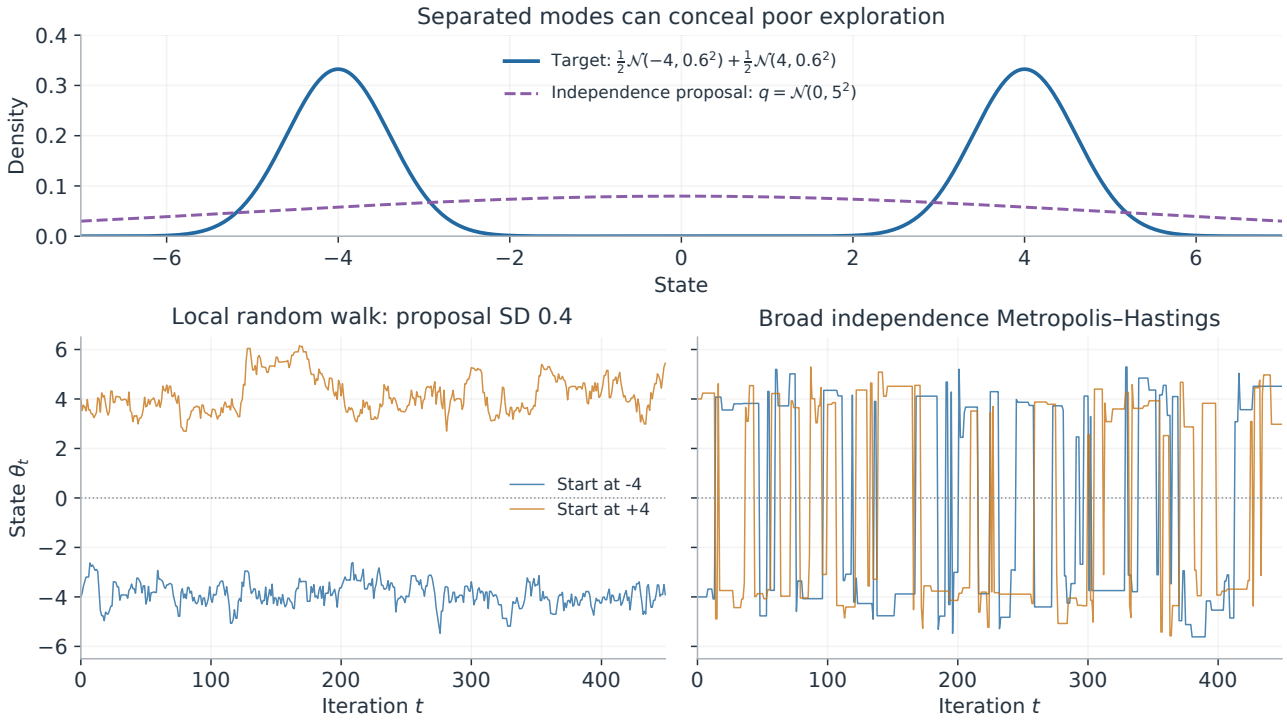


Figure 6: Within-mode exploration can look satisfactory while full-mixture exploration fails. Balanced starts can produce a plausible pooled histogram or mean without mode exchange. A broader proposal can improve mode travel despite accepting fewer candidates.

```
def mh_log_accept(lp_current, lp_candidate,
                  lq_forward, lq_reverse):
    return np.minimum(
        0.0, lp_candidate - lp_current + lq_reverse - lq_forward)
```

Forward means $q(y \mid x)$ and reverse means $q(x \mid y)$. For the independence proposal these are $q(y)$ and $q(x)$:

```
lq_forward = norm.logpdf(candidate, loc=0.0, scale=5.0)
lq_reverse = norm.logpdf(current, loc=0.0, scale=5.0)
log_alpha = mh_log_accept(lp_current, lp_candidate,
                          lq_forward, lq_reverse)
```

Independence from the current location does not imply conditional-proposal symmetry. Although this proposal is symmetric around zero, generally $q(x) \neq q(y)$. The Hastings terms do not cancel. For the Gaussian random walk, $q(y \mid x) = q(x \mid y)$ and they do cancel.

A useful hint is to write the ratio with explicit arguments before taking logs. A common additive constant in target log densities must cancel. Equal proposal terms must recover the symmetric rule. Checking detailed balance for one asymmetric pair directly verifies the orientation.

A.3 The law obtained when the correction is omitted

Suppose an independence sampler incorrectly uses $\alpha_0(x, y) = \min\{1, \pi_{\text{mix}}(y)/\pi_{\text{mix}}(x)\}$. Its stationary density is proportional to

$$\gamma(x) = \pi_{\text{mix}}(x)q(x).$$

The accepted flow under γ is

$$\gamma(x)q(y)\alpha_0(x, y) = q(x)q(y)\min\{\pi_{\text{mix}}(x), \pi_{\text{mix}}(y)\},$$

which is symmetric. Equivalently, the correct independence-MH ratio for γ cancels the proposal factors and becomes the incorrect target-only ratio.

A longer run therefore reduces numerical error around the wrong distribution. Here both π_{mix} and q are symmetric around zero. Their normalized product is symmetric too: even the wrong law has mean zero and positive-state probability one half. Those summaries cannot detect the missing correction. Algebraic validity and richer distributional checks are necessary.

A.4 Mode occupancy, transitions, and model answers

The local random walk typically explores its initial component with few or no inter-mode transitions. Starting two chains in each mode balances the pooled mean near zero by construction. It does not mean any individual chain has sampled the full mixture. Each chain's positive-state fraction can remain near zero or one, and split $\widehat{R}$ exposes substantial between-chain disagreement relative to component variance.

Count sign changes between consecutive states of each retained chain and report its positive-state fraction. These address mode exchange more directly than acceptance alone. A smooth trace in one component shows local movement, not global exploration. A batch MCSE calculated entirely inside one component can be small while omitting uncertainty associated with the other.

The broad independence proposal can move directly between modes. It wastes some candidates in low-density locations while improving global exploration. Acceptance, mode travel, and precision per evaluation measure different aspects of performance. No particular transition count should be required from every realization.

The task is complete when the proposal terms have the correct orientation, the formula reduces to the symmetric rule where appropriate, and the interpretation separates local movement from full-mixture exploration. A correct pooled mean from nonmixing chains is a counterexample, not a convergence certificate.

References

[1] Art B. Owen. *Monte Carlo theory, methods and examples*, Chapter 9: Importance sampling. Available from the author's website.

[2] Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2):667–718, 2021. arXiv:1903.08008.